

More Induction

WTP $\forall n \in \mathbb{Z}_{\geq 0}, P(n)$

BC: Prove $P(1)$.

IS: Prove " $P(a) \Rightarrow P(a+1)$ "

Prove: $\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Proof: Proceed by induction.

Base case: $P(1) \Rightarrow 1^2 = 1(2)(3)/6$

$$FE \quad 1 = 1.$$

Inductive step: Assume $P(a)$, i.e.,

$$1^2 + 2^2 + \dots + a^2 = a(a+1)(2a+1)/6. \quad \left(\begin{array}{l} \text{WTP} \\ P(a) \Rightarrow P(a+1) \end{array} \right)$$

Add $(a+1)^2$ gives

$$1^2 + 2^2 + \dots + a^2 + (a+1)^2 = \frac{a(a+1)(2a+1)}{6} + (a+1)^2.$$

The LHS is the LHS of $P(a+1)$.

$$\text{The RHS is } a(a+1)(2a+1) + (a+1)^2 =$$

$$(a+1) \left(\frac{a(2a+1)}{6} + \frac{6}{6} (a+1) \right) =$$

$$(a+1) \left(\frac{2a^2 + a + 6a + 6}{6} \right) = (a+1) \left(\frac{2a^2 + 7a + 6}{6} \right)$$

$$\frac{(a+1)(a+2)(2a+3)}{6} =$$

$$\frac{(a+1)(a+2)(2(a+1)+1)}{6}, \quad \text{This is the}$$

RHS of $f(a+1)$. $\square$

Claim: $\forall a \in \mathbb{Z}_{\geq 0}, 3^a$ is odd.

Proof BC: $P(0)$ is " 3^0 is odd",
i.e., 1 is odd. This is true.

FS: Assume $P(a)$, i.e., 3^a is odd.

(WTP: $P(a+1)$, i.e., 3^{a+1} is odd)

$$\boxed{3^a} \cdot 3$$

Then $3^{a+1} = 3^a \cdot 3$. Since 3^0 is odd
+ 3 is odd, and since the
product of odd integers is odd,

3^{a+1} is odd. ~~QED~~

Slogan: "Any time something 'works' for
2 things, it works for many things."

Example: we know that "odd · odd = odd"

Pf: $(2a+1) \cdot (2b+1) = 4ab + 2(a+b) + 1$
rem-odd. $\rightarrow$

Claim: If $a_1, \dots, a_n$ are odd integers
then $a_1 \cdot a_2 \cdot \dots \cdot a_n$ is odd! ? Gal

Pf: Proceed by induction.

BC is $P(1)$ i.e., "if a_1 is odd,
then a_1 is odd". This is true.

$P(2)$ is "if a_1, a_2 are odd, then
 a_1, a_2 is odd." This is true
and we previously proved it,

FS: Prove that for $n \geq 2$, $P(n) \Rightarrow P(n+1)$

Assume $P(n)$, i.e., "if $a_1, \dots, a_n$ are odd
then $a_1 \dots a_n$ is odd."

(WIP: $P(n+1) \nexists$ [if $a_1, \dots, a_{n+1}$ are odd] the
 $a_1 \dots a_{n+1}$ is odd]

Assume $a_1, \dots, a_{n+1}$ are odd.

Then $a_1, \dots, a_n$ is odd because $P(n)$ is true. Then $(a_1, \dots, a_{n+1}) = (a_1, \dots, a_n) \cdot a_{n+1}$.

Since $(a_1, \dots, a_n)$ is odd and a_{n+1} is odd,

Since we knew $P(n)$, their product is

odd. Thus $a_1, \dots, a_{n+1}$ is odd 

$$a+b = b+a$$

$$\begin{aligned} a+b+c &= a+(b+c) = a+(c+b) \\ &= (c+b)+a \end{aligned}$$

2 base cases

$$P(1) \wedge P(2) \wedge (P(n) \Rightarrow P(n+1))$$

The proof that $P(n) \Rightarrow P(n+1)$
doesn't work for $n=1$.

Prove; $n! > 2^n$ for $n \geq 4$.

$P(n)$

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 1$$

$n =$
1
2
3
4

$$\begin{aligned} P(n) = \\ 1! = 1 > 2 \\ 2! = 2 > 4 \\ 3! = 6 > 8 \\ 4! = 24 > 16 \end{aligned}$$

Proof, Proceed by induction.

BC: $P(4)$ is " $4! > 2^4$ ", i.e., " $24 > 16$ ".

This is true.

Assume $n \geq 4$ and assume $P(n)$, i.e., $n! > 2^n$.

(WTP $P(n+1)$, i.e., $(n+1)! > 2^{n+1}$)

Multiplying $P(n)$ by $n+1$ gives

$$(n+1)n! \geq (n+1)2^n.$$

The RHS of this is $(n+1)!$.

Since $n \geq 2$, $n+1 \geq 2 \geq 2$.

Thus

$$(n+1)2^n \geq 2 - 2^n = 2^{n+1}.$$

We conclude that $(n+1)! \geq 2^{n+1}$ $\square$

Fibonacci #'s

$$F_1 = 1, F_2 = 1, F_3 = 1 + 1$$

$$F_4 = 3, \dots$$

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

1st
Perfect Pair

Defn:

$$F_1 = 1, F_2 = 1$$

$$F_n = F_{n-1} + F_{n-2}$$

~~Same as~~

$$F_{n+1} = F_n + F_{n-1}$$

$$F_a = F_{a-1} + F_{a-2}$$

$$F_{n+2} = F_{n+1} + F_n$$

$$F_{a+2} = F_{a+1} + F_a$$

Claim: " $F_1 + F_3 + F_5 + \dots + F_{n-1} = F_n$ "

Ex's 1, 1, 2, 3, 5, 8, 13, 21 "P(n)"

Proof: (BC) P(1) is " $F_1 = F_2$ ", i.e.,

" $1 = 1$ ". This is true.

IS: Assume $P(n)$, i.e.,
" $F_1 + F_3 + \dots + F_{n-1} = F_n$ " $\quad \quad \quad \begin{array}{l} 2(n+1)-1 = \\ 2n+2-1 \\ 2n+1 \end{array}$

Adding F_{n+1} to both sides gives

$$F_1 + F_3 + \dots + F_{n-1} + F_{n+1} = F_n + F_{n+1}.$$

By the defn, $F_n + F_{n+1} = F_{n+2}$.

Thus $F_1 + F_3 + \dots + F_{2n-1} = F_{2n}$, i.e.,

$P(n+1)$ is true. $\square$

^u
"Lucas"

2, 1, 3, 4, 7, 11, 18, 29, ...

$$L_1 = 2 \quad L_2 = 1$$

$$L_n = L_{n-1} + L_{n-2}$$

Claim: $\forall n \in \mathbb{Z}_{>0}, \quad 'F_n < a^n'$

Proof: (BC) $P(1)$ is " $F_1 < a$ ", i.e.,
 $1 < a$. $P(a)$ is " $F_2 < a^2$ ", i.e.,

$1 < 4$. These are true.

IS: Assume $P(n)$ and $P(n+1)$, IE,

$$F_n < 2^n \text{ and } F_{n+1} < 2^{n+1}.$$

$$(\text{wTP } P(n+2), \text{ i.e., } F_{n+2} < 2^{n+2})$$

By defn. $F_{n+2} = F_{n+1} + F_n$. Since $P(n)$ and

$$P(n+1) \text{ are true, } F_{n+1} + F_n < 2^{n+1} + 2^n,$$

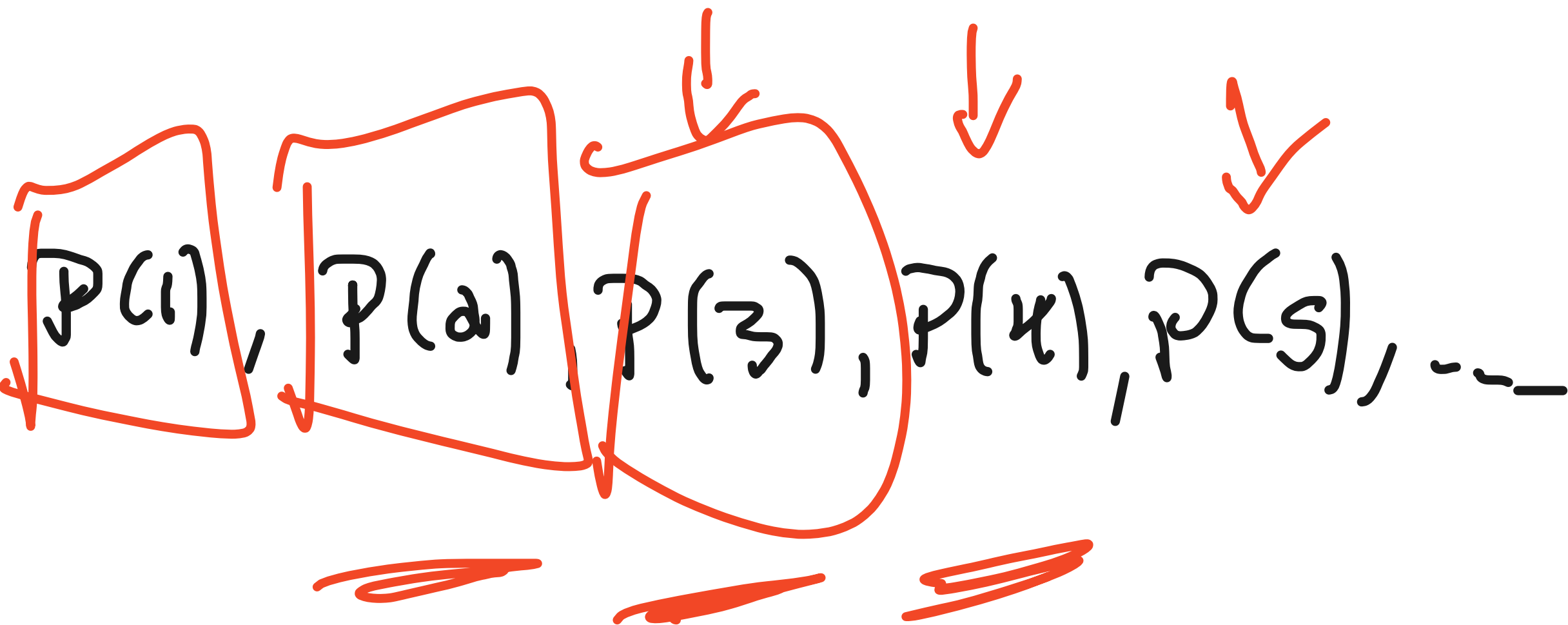
$$\text{I.E., } F_{n+a} < 2^{n+1} + 2^n. \quad (\text{want } 2^{n+2})$$

$$\text{Since } 2^n < 2^{n+1}, \quad 2^{n+1} + 2^n < 2^{n+1} + 2^{n+1}$$

$$= 2 \cdot 2^{n+1} = 2^{n+2}. \quad \text{Thus}$$

$$F_{n+a} < 2^{n+2}. \quad \square$$

$$P(1) \wedge P(2) \wedge (P(n) \wedge P(n+1)) \Rightarrow P(n+1)$$



Claim: " $F_{n-1} \cdot F_{n+1} = F_n^2 + (-1)^n$ " = $P(n)$

Proof: " $P(2)$ " is the statement (1, 1, 2, 3, 5, 8)

$$"F_1 \cdot F_3 = F_2^2 + (-1)^2"$$

$$1 \cdot 2 = 1 + (-1)^2$$

$2 = 2$. This is true.

FS - Assume $P(a), i.e., F_{a-1} \cdot F_{a+1} = F_a^2 + (-1)^a$.
(wtp $P(a+1), i.e., F_a \cdot F_{a+2} = F_{a+1}^2 + (-1)^{a+1}$)

By defn, $F_{a-1} + F_a = F_{a+1}, i.e., F_{a-1} = F_{a+1} - F_a$.

Substing gives $(F_{a+1} - F_a) \cdot F_{a+1} = F_a^2 + (-1)^a$.

Then $F_{a+1}^2 - F_a \cdot F_{a+1} = F_a^2 + (-1)^a$.

Then $F_{a+1}^2 - (-1)^a = F_a^2 + F_a \cdot F_{a+1},$

Then $F_{a+1}^2 + (-1)(-1)^a = F_{a+1}^2 + (-1)^{a+1}$

$= F_a(F_a + F_{a+1}) = F_a F_{a+2}$ by the def. $\square$

def of F.b.

